CHEW SIN AI

SOFTWARE ENGINEER

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SUMMARY

Software Engineer with hands-on enterprise experience in full-stack web, mobile, and workflow automation. Specialized in developing Flutter warehouse scanner applications with offline-first caching, core Java WMS modules, automated PHP cron transfer/billing jobs, and Gemini AI document extraction pipelines. InIIC 2024 Bronze Medalist & published researcher.

EXPERIENCE

Tiong Nam Logistics | Software Engineer — **Oct 2025 – Present**

- • **Flutter Scanner App:** Built ruggedized Android warehouse scanner app in Flutter with offline SQLite caching and hardware scan triggers (Inbound, Outbound, PSC audit).
- • **Enterprise WMS Web:** Maintained core Java enterprise WMS platform; developed inventory location transfers and resolved concurrency bottlenecks in MySQL.
- • **Cron & Billing Automation:** Engineered automated PHP cron jobs for daily stock transfers, billing calculations, and email dispatches with file-lock concurrency guards.

Assignment Helper | CS Academic Tutor (Part-Time) — **2025 – Present**

- • **Mentorship:** Tutored undergraduate students in Java, Python, Object-Oriented Programming (OOP), Data Structures & Algorithms, and code debugging.

Rockbell Software Company | Software Support Engineer (Part-Time) — **Jun 2025 – Jul 2025**

- • **ERP Support:** Provided technical support and executed diagnostic SQL queries to troubleshoot customer ERP database and data integrity issues.

SELECTED PROJECTS

Patient Care Real-Time Management System | Laravel, Flutter, Firebase, MySQL — **2024**

- Sole developer of a hospital communication platform featuring a 4-role Laravel web portal and Flutter mobile app with real-time Firebase sync. Won Bronze Award at InIIC 2024 with published research thesis.

WMS Handheld Scanner Mobile Application | Flutter, SQLite, REST API [Enterprise / NDA] — **2025 – 2026**

- Engineered production Android scanner for floor logistics operations, reducing manual barcode logging through instant hardware trigger integration and offline-resilient local sync.

AI Mail & Document Ingestion Pipeline | React, PHP, Gemini AI, Playwright [Enterprise / NDA] — **2025 – 2026**

- Automated operational document intake: parses incoming shipment emails, extracts fields via Gemini multimodal AI OCR, and submits directly into enterprise portals via Playwright.

AI PDF Split, Merge & Renaming Engine | FastAPI, React, Gemini AI, PyMuPDF [Enterprise / NDA] — **2025 – 2026**

- High-throughput document intelligence service that splits bulk invoices/DOs, performs OCR extraction with Gemini AI, and renames/sorts files automatically according to client rules.

TECHNICAL SKILLS

Languages: PHP, Java, Dart, Python, JavaScript, HTML5/CSS3, C++, SQL
Frameworks & Mobile: Flutter, Laravel, FastAPI, React.js, JavaFX, PyMuPDF
Databases & Cloud: MySQL, SQLite, Firebase (Realtime DB, Firestore, Auth, FCM), REST APIs
Tools & Automation: Gemini AI, Playwright, Linux / Cron Jobs, Git, Android Studio

EDUCATION & CERTIFICATIONS

Bachelor of Software Engineering (Honours) | Southern University College — **2022 – 2025**

- CGPA: 3.70 / 4.00 · Dean's List Award Recipient · InIIC 2024 Bronze Award & Publication

Certifications & Languages: NVIDIA CUDA Accelerated Computing | English (Professional), Chinese (Native), Malay (Professional)